

PHISHING DETECTION & AWARENESS REPORT

Email Analysis • Risk Classification • User Awareness

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Purpose

This report demonstrates a structured, non-invasive approach to identifying phishing indicators in email messages and converting those findings into practical security-awareness guidance. No live phishing links were intentionally opened or executed.

Classification scale: Safe • Suspicious • Phishing

1. Executive Summary

Phishing is a social-engineering technique in which an attacker impersonates a trusted person, service or organization to persuade a recipient to disclose information, click a malicious link, open an unsafe attachment or perform another harmful action.

For this task, the analysis focuses on the observable characteristics of phishing messages: sender identity, urgency, fear-based language, suspicious URLs, generic greetings, requests for sensitive information and the overall consistency of the message with a legitimate security notification.

Key findings

- **Urgency is a recurring manipulation technique.** Deadlines such as “within 24 hours” pressure recipients into acting before verifying the request.
- **Threats and fear increase compliance.** Account-lock or suspension language makes the recipient more likely to click without checking.
- **Sender and domain verification is essential.** A familiar display name is not proof that the underlying email address is legitimate.
- **URL inspection provides an important warning signal.** The actual domain should be checked before any credentials or financial information are entered.
- **Awareness must complement technical controls.** Users should know how to report suspicious messages and how to verify requests through trusted channels.

Overall assessment

Metric	Result	Interpretation
Email scenarios reviewed	3	One task-provided example + two sanitized representative training scenarios
High-risk classifications	3 / 3	All scenarios contain multiple phishing indicators
Primary attack goal	Credential / information theft	Victim is pushed toward a fake verification or login workflow

OVERALL: HIGH RISK

2. Scope, Objectives & Methodology

The task is explicitly positioned as security education and analysis rather than hacking. Accordingly, the assessment is designed to be safe, reproducible and non-invasive.

Objectives

- Analyze phishing email examples and describe the indicators that make them suspicious.
- Inspect sender/domain and URL characteristics without intentionally visiting malicious destinations.
- Use header-analysis concepts such as From, Reply-To, Return-Path, Received, SPF, DKIM and DMARC where header data is available.
- Classify messages as Safe, Suspicious or Phishing using multiple indicators rather than a single clue.
- Explain the attack in simple, non-technical language and provide practical employee guidance.

Analysis workflow

1. Collect	2. Inspect	3. Correlate	4. Classify	5. Educate
Obtain safe samples and redact sensitive data	Review sender, subject, body, URLs and available headers	Compare indicators and check consistency	Assign Safe / Suspicious / Phishing	Document prevention and reporting steps

Evidence-handling note

No live malicious link was opened. Suspicious URLs in examples are treated as text indicators and are redacted or represented with placeholders where appropriate. This avoids turning a defensive assessment into interaction with potentially harmful infrastructure.

Important limitation

The task-provided example is a mock phishing email and does not include a full raw header. Therefore, this report does not invent SPF/DKIM/DMARC or IP-level results. Header findings are reported only when evidence is actually available.

3. Sample 01 — Account Lock / Verification Phishing

Source type: Task-provided sample email from the Future Interns assignment.

Subject: ■■ Urgent: Your Account Will Be Locked

Sanitized message excerpt

Dear User,

We noticed suspicious activity on your account.
To avoid account suspension, please verify your details immediately.

Verify Now: [SUSPICIOUS URL REDACTED]

Failure to verify within 24 hours will result in permanent account lock.

Regards,
Security Team

Indicator analysis

Indicator	Observation	Assessment
Generic greeting	The message addresses the recipient as “Dear User” rather than using a verified customer identity.	High
Urgency	The recipient is told to act “immediately” and within 24 hours.	High
Fear / threat	Permanent account lock is presented as the consequence of not complying.	High
Credential-verification request	The recipient is directed to verify account details.	Critical
Suspicious link	The visible URL is represented as an untrusted verification destination.	Critical
Sender identity	“Security Team” is generic and does not establish a trusted organization.	Medium/High

Risk classification

PHISHING — HIGH RISK

How the attack works

The attacker creates fear of account loss, provides a short deadline and then offers a verification link as the supposed solution. A victim who follows the link may be taken to a fraudulent page designed to collect login or other sensitive information.

Header assessment

No raw header was supplied with this task-provided example. Therefore, SPF/DKIM/DMARC and Received-path results are **not claimed**. A real investigation should obtain the original message headers and verify those fields.

4. Sample 02 — Banking / Transaction Verification

Source type: **Sanitized representative training scenario** based on common phishing patterns. It is not presented as a live malicious email.

Subject: **Action Required: Unusual Transaction Detected**

Sanitized message excerpt

Dear Customer,

We detected an unusual transaction on your account. If you do not confirm your identity, your account may be temporarily restricted.

Review Transaction: [SUSPICIOUS URL REDACTED]

Please complete verification as soon as possible.

Indicator analysis

Indicator	Observation	Assessment
Financial theme	The message uses a transaction alert to create concern and prompt immediate action.	High
Impersonation	"Dear Customer" gives no trustworthy evidence of the actual bank or payment provider.	Medium
Urgency	The recipient is encouraged to verify as soon as possible.	High
Account restriction threat	A possible restriction is used to increase pressure.	High
Verification link	The user is directed away from a trusted banking workflow toward an unverified destination.	Critical
Sensitive-information risk	A fake verification page could request credentials or financial information.	Critical

Risk classification

PHISHING — CRITICAL RISK

Why this is dangerous

Financial-themed phishing messages exploit the recipient's fear of unauthorized transactions. The safest response is to open the bank's official app or type the known official website address manually, rather than using the email link.

Safe verification procedure

- Do not use the email's link to access the account.
- Check the transaction through the official banking app or a trusted bookmark.
- Contact the institution using a verified phone number or official support channel if the transaction is real.

5. Sample 03 — Password Expiration / Login Phishing

Source type: **Sanitized representative training scenario** based on common account-login phishing patterns. It is not presented as a live malicious email.

Subject: **Your Password Expires Today — Sign In to Continue**

Sanitized message excerpt

Hello,

Your password is scheduled to expire today. To keep your mailbox active, sign in and confirm your account.

Keep My Account Active: [SUSPICIOUS URL REDACTED]

Failure to complete verification may affect access to your account.

Indicator analysis

Indicator	Observation	Assessment
Password-expiration lure	The message creates a routine administrative problem that appears urgent.	High
Account-access threat	The recipient is warned that access may be affected.	High
Login request	The recipient is directed toward a sign-in workflow.	Critical
Unverified destination	The login link is represented as suspicious rather than as a trusted corporate domain.	Critical
Generic wording	The message lacks strong evidence connecting it to the recipient's actual service provider.	Medium
Credential-harvesting potential	A fake sign-in page could capture usernames and passwords.	Critical

Risk classification

PHISHING — HIGH RISK

Key lesson

Password-reset and expiration messages should be verified independently. Users should access the service through a known official bookmark, company portal or manually entered domain instead of using an unexpected email link.

Header assessment

Because no raw header is included in this sanitized scenario, technical authentication results are not asserted. In a real investigation, the original header would be required before making sender-authentication conclusions.

6. Comparative Findings & Risk Classification

The table below compares the observable indicators across the three scenarios. The classification is based on the combination and severity of indicators, not on any single keyword.

Indicator	Sample 01	Sample 02	Sample 03
Urgency / deadline	✓	✓	✓
Fear / account restriction	✓	✓	✓
Generic greeting	✓	✓	✓
Suspicious verification/login link	✓	✓	✓
Credential / sensitive-data risk	✓	✓	✓
Social engineering	Urgency + fear	Financial concern	Access concern
Final classification	PHISHING	PHISHING	PHISHING

Risk classification framework

SAFE	SUSPICIOUS	PHISHING
No meaningful indicators; sender, context and destination are consistent with a legitimate message.	One or more anomalies exist, but evidence is insufficient for a confident phishing classification.	Multiple strong indicators or clear deceptive intent are present, especially credential or sensitive-data harvesting.

Analyst conclusion

All three reviewed scenarios should be treated as phishing/high risk because they combine social pressure with an unverified link or login/verification workflow. The safest user action is to stop, verify independently and report the message.

7. Technical Indicators: Headers, Domains & URLs

A complete phishing investigation should combine message content with technical evidence. The following checklist can be applied to a real email without opening suspicious content.

Email-header checklist

Field	What to examine	Why it matters
From	Actual sender address and domain	Display names can be spoofed or misleading.
Reply-To	Whether replies are redirected to another domain	A mismatch can be a warning sign.
Return-Path	Envelope sender	Can reveal unexpected sending infrastructure.
Received	Mail-transfer path and timestamps	Helps reconstruct the message route.
SPF	Whether the sending server is authorized	Supports sender-domain validation.
DKIM	Whether the message has a valid signature	Helps verify message integrity and domain alignment.
DMARC	Policy/alignment result	Combines authentication signals and domain alignment.

URL/domain checklist

- Read the registrable domain, not just the visible brand words.
- Look for misspellings, unexpected subdomains, unusual top-level domains and shortened URLs.
- Treat HTTP links, IP-address links and unexpected login pages with caution.
- Do not assume HTTPS means the site is trustworthy; HTTPS protects the connection, not the legitimacy of the website.
- If verification is needed, navigate independently to the organization's known official website or app.

Tool references

The task recommends Google Admin Toolbox Messageheader and MXToolbox Email Header Analyzer for header inspection. These tools are useful for understanding routing and authentication fields, but their results should be interpreted in context.

8. Phishing Prevention & Employee Awareness

The goal of awareness training is to give employees a repeatable decision process. A user does not need to become a security expert to stop a phishing attempt.

The **STOP** → **CHECK** → **VERIFY** → **REPORT** model

STOP	CHECK	VERIFY	REPORT
Pause when an email creates urgency, fear or pressure.	Inspect sender, domain, spelling, URL and request.	Use a trusted app, bookmark, official website or known contact method.	Report the message to the organization's IT/security process and then remove it as instructed.

Employee Do's

- Do verify unusual requests independently.
- Do inspect the actual sender address and link destination.
- Do use multi-factor authentication where available.
- Do report suspected phishing promptly.
- Do ask the security/IT team when a request feels unusual.

Employee Don'ts

- Don't provide passwords, OTPs or payment information through an unexpected email.
- Don't click a link simply because the email looks professional.
- Don't rely on logos, signatures or display names as proof of authenticity.
- Don't forward suspicious links to colleagues unnecessarily.
- Don't ignore a suspected compromise after entering credentials.

9. What To Do If a User Clicked

A good awareness report should also explain the response after a mistake. The objective is to reduce damage quickly and avoid blame, because fast reporting can improve the organization's ability to contain an incident.

Situation	Immediate action
Clicked the link only	Close the page, do not enter information, report the message and follow the organization's security guidance.
Entered a password	From a trusted device/channel, change the password immediately. If reused elsewhere, change it there too. Notify IT/security.
Entered OTP / authentication code	Report immediately. Follow the organization's account-security procedure and review active sessions if available.
Entered banking/payment details	Contact the bank/payment provider using an official number or app immediately and follow its fraud procedure.
Downloaded a suspicious file	Do not open it. Disconnect from networks if instructed by security/IT and report the incident for investigation.

Incident-report checklist

- Preserve the original email if the security team requests it.
- Record the approximate time of the event.
- Tell the security/IT team exactly what was clicked, downloaded or entered.
- Do not delete evidence before the security team has had a chance to collect it, unless instructed otherwise.

Core principle

Report quickly, even if you are unsure. Early reporting is more useful than hiding a mistake.

10. Conclusion & Recommendations

The assessment shows how phishing messages combine technical deception with social engineering. The strongest warning signs are not limited to spelling mistakes; attackers can produce polished messages that still contain a dangerous request or untrusted destination.

Recommendations for organizations

- **Strengthen email authentication:** maintain SPF, DKIM and DMARC correctly and monitor authentication failures.
- **Provide recurring awareness training:** teach employees how to recognize urgency, impersonation, suspicious links and credential requests.
- **Make reporting easy:** provide a visible phishing-report mechanism and define a clear response process.
- **Use technical controls:** filtering, link analysis, attachment scanning, browser protections and MFA should complement user awareness.
- **Measure improvement:** track reporting rates, repeat-risk patterns and training outcomes rather than only counting blocked emails.

Final assessment



PHISHING
AWARENESS:
PRIORITY CONTROL

The three scenarios reviewed in this report are classified as phishing because they use a combination of urgency, fear, impersonation-style language and unverified verification/login workflows. Users should stop, inspect, independently verify and report rather than interact with the message.

Limitations & ethics

- Two scenarios are explicitly sanitized representative training examples and are not claimed to be live malicious emails.
- No credentials, OTPs or private personal information were collected.
- No suspicious URL was intentionally opened or executed.
- Technical header conclusions are not fabricated where raw header evidence is unavailable.

11. References, Tools & Evidence Notes

Public study/reference sources

1. Phishing.Database — public phishing-domain, URL and threat-intelligence project. The repository describes itself as a regularly updated resource for identifying and mitigating phishing threats.
2. Phishing.Database contribution repository — documents how phishing domains, links and IPs are contributed and reviewed.
3. Google Admin Toolbox Messageheader — referenced by the assignment for email-header analysis.
4. MXToolbox Email Header Analyzer — referenced by the assignment for email-header analysis.

Links

Phishing.Database: <https://github.com/Phishing-Database/Phishing.Database>

Phishing.Database contribution hub: <https://github.com/Phishing-Database/phishing>

Google Admin Toolbox Messageheader: <https://toolbox.googleapps.com/apps/messageheader/>

MXToolbox Email Header Analyzer: <https://mxtoolbox.com/EmailHeaders.aspx>

Evidence and attribution note

The task-provided account-lock example is reproduced only in sanitized form for analysis. The remaining scenarios are written as representative training examples so that the report demonstrates the requested analysis process without copying or redistributing live malicious content. Public repositories are cited for study/reference purposes and should not be represented as the author's original content.

Suggested GitHub repository structure

```
phishing-email-detection-awareness/  
■■■ README.md  
■■■ report/Phishing_Detection_Awareness_Report.pdf  
■■■ samples/  
■ ■■■ phishing_sample_01.txt  
■ ■■■ phishing_sample_02.txt  
■ ■■■ phishing_sample_03.txt  
■■■ evidence/  
■■■ dashboard/
```

End of report